



Complex Systems

2020/2021

Exam R 02 of July 2021 Time: 2h00

Name:

Number:

Question #	Points	Your score
1	18	
2	12	
3	25	
4	25	
5	20	
	100	

Grader:

Pergunta 1 - Define 18 points

Define the following concepts in a clear and **rigorous** way. You may use a concrete example to **help you** to explain the concept.

a) Basin of Attraction.

Your answer:

b) Constant of Feingenbaum.

Your answer:

c) Chaos Game.

Your answer:

Pergunta 2 - Comment **12 points**

Comment the following statements. **Remember**, to comment is not just to say agree/disagree for you have to justify your evaluation.

- a) There is a relationship between the bifurcation diagram and the Lyapunov exponent.

Your answer:

Your answer:

b) All first-order discrete linear system can be solved algebraically.

Your answer:

Your answer:

Pergunta 3 - Architecture 25 points

Consider a 1D cellular automata with rectangular symmetry and neighborhood of radius 1. Suppose that each cell can be in one of three states : $\oplus$, $\otimes$, $\odot$. Suppose further that the transition of one cell's state depend only on the state of its two neighborhoods (left and right).

a) How many transition rules are possible?👉

Your answer:

- b) Suppose that the transition rule is as follow:
- if the two neighborhoods cells have the same state value then the next value of the cell will be also the same;
 - If they are different they can be ordered by priority according with $\oplus > \otimes > \odot$.

Under these conditions how would evolve the AC of the figure 1?

$\otimes$	$\otimes$	$\otimes$	$\oplus$	$\oplus$	$\oplus$	$\odot$	$\odot$	$\odot$

Figure 1: Configuração inicial.

Your answer:

Pergunta 4 - Explain 25 points

Consider the system described by the differential equation:

$$\frac{dx}{dt} = x$$

Can this systems present a **chaotic** behavior? Justify with rigor.

Your answer:

Your answer:

Pergunta 5 - Solve 20 points

Imagine an ecosystem where there are two species that **instead of competing cooperate**. Moreover, even if they can survive alone in the environment, if they are together they can develop better. We say that the two species live in **symbiosis**. Model the system by means of differential equations, explaining clearly your options.

Your answer:

Your answer: